

SAKSHAM KUMAR

Software Engineering Student | Full-Stack MERN Developer

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TECHNICAL SKILLS

Languages: JavaScript (ES6+), Java, SQL

Frontend: React.js, Tailwind CSS, Vite

Backend: Node.js, Express.js, REST API design, WebSockets (Socket.io), JWT / Auth flows

Databases: MongoDB (document modelling), MySQL (relational design, query optimisation)

Infra & Tooling: Docker, Redis, Git, GitHub, Postman

CS Core: Data Structures & Algorithms, OOP, System Design (in progress)

EXPERIENCE

Full-Stack Developer Intern — InstanDev (Remote)

Jun 2026 – Aug 2026 (Expected)

InstanDev Internship Cohort 2026

- Built **WORDLE.BATTLE**, a real-time multiplayer word-guessing game, architecting both the Node.js/Express backend and the React 19 frontend — supporting random matchmaking and host-controlled private lobbies over a single Socket.IO connection.
- Designed the auth layer with short-lived JWT access tokens and rotating refresh tokens stored in httpOnly cookies, zod-validated request schemas, and bcrypt password hashing.
- Added a Socket.IO Redis adapter and per-IP rate limiting for horizontal scalability, and containerized the frontend, backend, MongoDB, and Redis with Docker Compose for reproducible deployment.

PROJECTS

DOXY — Telemedicine Platform

Node.js · Socket.io · WebRTC · REST APIs | Smart India Hackathon 2025

- Architected a backend sustaining 1,000+ concurrent users under Artillery load tests; implemented WebRTC peer connections with Socket.io as the signalling layer for patient–doctor video sessions.
- Designed the async request pipeline for a third-party NLP triage API — including timeout handling, error boundaries, and response normalisation before surfacing results to the client.

Chatifyy — Real-Time Messaging Platform

MERN Stack · Socket.io · JWT

- Shipped a MERN-based chat system with Socket.io event-driven messaging, supporting concurrent group and private rooms with live user presence tracked via connection state.
- Secured all REST routes and WebSocket handshakes with JWT verification and role-based middleware (admin/user), keeping auth logic decoupled from business logic.

ACHIEVEMENTS & INTERESTS

- Smart India Hackathon 2025 — Sole backend architect on DOXY; system cleared national-level evaluation handling real-time video + concurrent user load.
- Solved 150+ LeetCode problems (graphs, DP, trees); studying system design patterns — consistent daily practice since Jan 2024.

EDUCATION & LANGUAGES

B.Tech in Computer Science and Engineering

Aug 2023 – May 2027 (Expected)

GLA University, Mathura, Uttar Pradesh

Languages: English (Professional Proficiency) • Hindi (Native)